

LAKSHYA TYAGI

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EDUCATION

Harcourt Butler Technical University

Bachelor of Technology in Information Technology (CGPA: 7.456 / 10.00)

Aug 2023 – May 2027

Kanpur, India

POSITION OF RESPONSIBILITY

Event Volunteer - ICPC 2025 Offline Prelims

HBTU (CSE Department)

Oct 2025 – Dec 2025

- Volunteered in organizing the 2025 ICPC Offline Prelims at HBTU (CSE Dept), managing logistics and coordination for 40+ participants.

PROJECTS

C++ Debug Header — Generic dbg Utility | C++, Template Metaprogramming, SFINAE, Type Traits (void_t, enable_if)

- Developed a lightweight C++ debug header to print STL containers, pairs, tuples, optionals, and user-defined types using template metaprogramming.
- Implemented a variadic dbg() macro with clean formatting for nested structures and a helper for runtime-sized arrays, improving debugging speed by 50–60% in competitive coding.
- Applied C++ templates, type traits, SFINAE, and STL utilities to design a flexible and dependency-free debugging tool.

Plant Disease Classification — PCA + CNN and ICA + CNN | Python, NumPy, Pandas, Matplotlib, TensorFlow

- Applied PCA and ICA with a custom CNN to classify four plant leaf classes on a PlantVillage-mini dataset of 5,766 RGB images (64×64); applied stratified splits, basic augmentation, and class balancing.
- Trained and compared PCA→CNN and ICA→CNN pipelines; ICA showed a modest accuracy improvement (90% to 93%) and improved Tomato Late-blight recall (0.75 to 0.85).
- Designed a training workflow with early stopping, learning-rate tuning, and evaluation using precision, recall, F1-score, and confusion matrices for clearer error analysis.

COMPETITIVE PROGRAMMING PROFILES

- Codeforces** – 1670 (Expert) (Max.)
- Codechef** – 1856 (4 star) (Max.)
- Atcoder** – 912 (Max.)
- Leetcode** – 1819 (Max.)
- ICPC** – 2 × ICPC Regional Finalist (2024 – 2025, 2025 – 2026)

TECHNICAL SKILLS

Programming Languages and Databases: C, C++, Python, SQL

Frameworks and Libraries: NumPy, Matplotlib, Pandas

Version Control: Git, GitHub

OS & Tools: Arch Linux, Windows, Vim, Google Colab, VS Code, Sublime Text

Other Skills: Data Structures and Algorithms, Object-Oriented Programming, Computer Networks, Operating Systems, DBMS

ACHIEVEMENTS

- Secured the **157th** rank out of **2,930** teams in the **ICPC Preliminary Contest 2025**.
- Secured the **195th** rank out of **2,429** teams in the **ICPC Preliminary Contest 2024**.
- Secured the **189th** rank in the **ICPC Asia Amritapuri Regional Contest 2024 – 25**.
- Secured **Rank 4099** in the **Meta Hacker Cup 2024** and qualified for **Round 2** among thousands of global participants.
- Solved **1,800+** problems across platforms including LeetCode, Codeforces, CodeChef, AtCoder and GeeksForGeeks.
- Secured **Global Rank 362** out of 10,000+ participants in **Codeforces Round 1058 (DIV 2)**.
- Secured **2nd Best Innovators Award** at **KodeKalesh'25 Devfolio Hackathon** (sponsored by AWS, GfG, EthIndia) among 2000+ participants.
- Qualified **Smart India Hackathon 2024** college-level round under the Heritage and Culture theme
- Achieved **97 percentile** in Joint Entrance Examination Mains (with **99.4 percentile in Mathematics**).
- Achieved **18031 AIR** in Joint Entrance Examination Advanced. Also qualified for prestigious entrance exams including **IAT** and **NEST**.

CERTIFICATION AND TRAINING

One-Week Summer Training Program — Skill Development

HBTU Kanpur

July 2024

- Completed certified training covering Python programming, OOP, NumPy/Pandas, basic AI/ML concepts, and soft-skill development through expert-led technical sessions.